Data Preprocessing and Mining (Lecture - 7)



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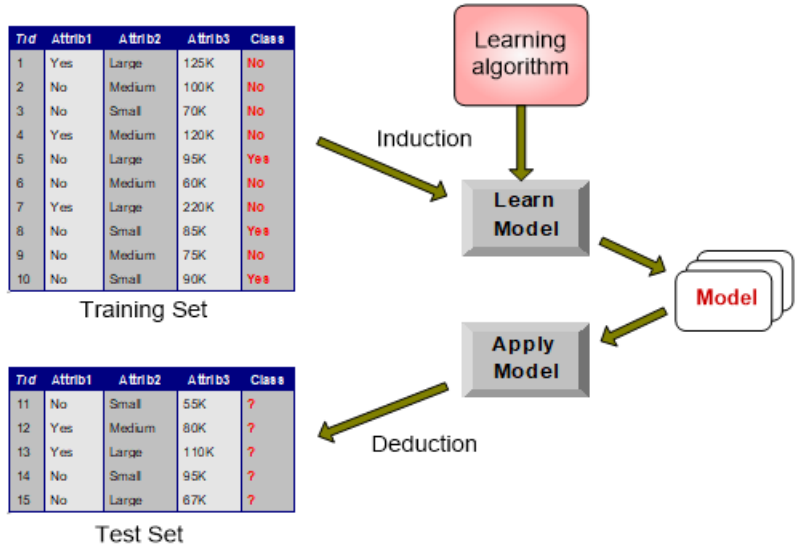
- Advantages:
  - Relatively inexpensive to construct
  - Extremely fast at classifying unknown records
  - Easy to interpret for small-sized trees
  - Robust to noise (especially when methods to avoid overfitting are employed)
  - Can easily handle redundant attributes
  - Can easily handle irrelevant attributes (unless the attributes are interacting)
- Disadvantages: .
  - Due to the greedy nature of splitting criterion, interacting attributes (that can distinguish between classes together but not individually) may be passed over in favor of other attributed that are less discriminating.
  - Each decision boundary involves only a single attribute

Decision Tree Based Classification

Training errors: Errors committed on the training set

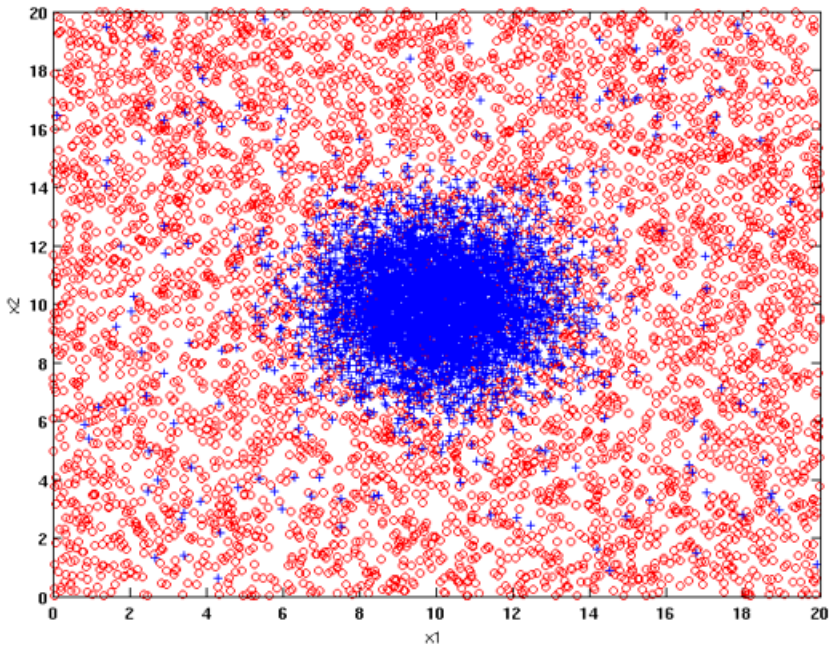
Test errors: Errors committed on the test set

Generalization errors: Expected error of a model over random selection of records from same distribution



Decision Tree Based Classification

Example Dataset



Two class problem:

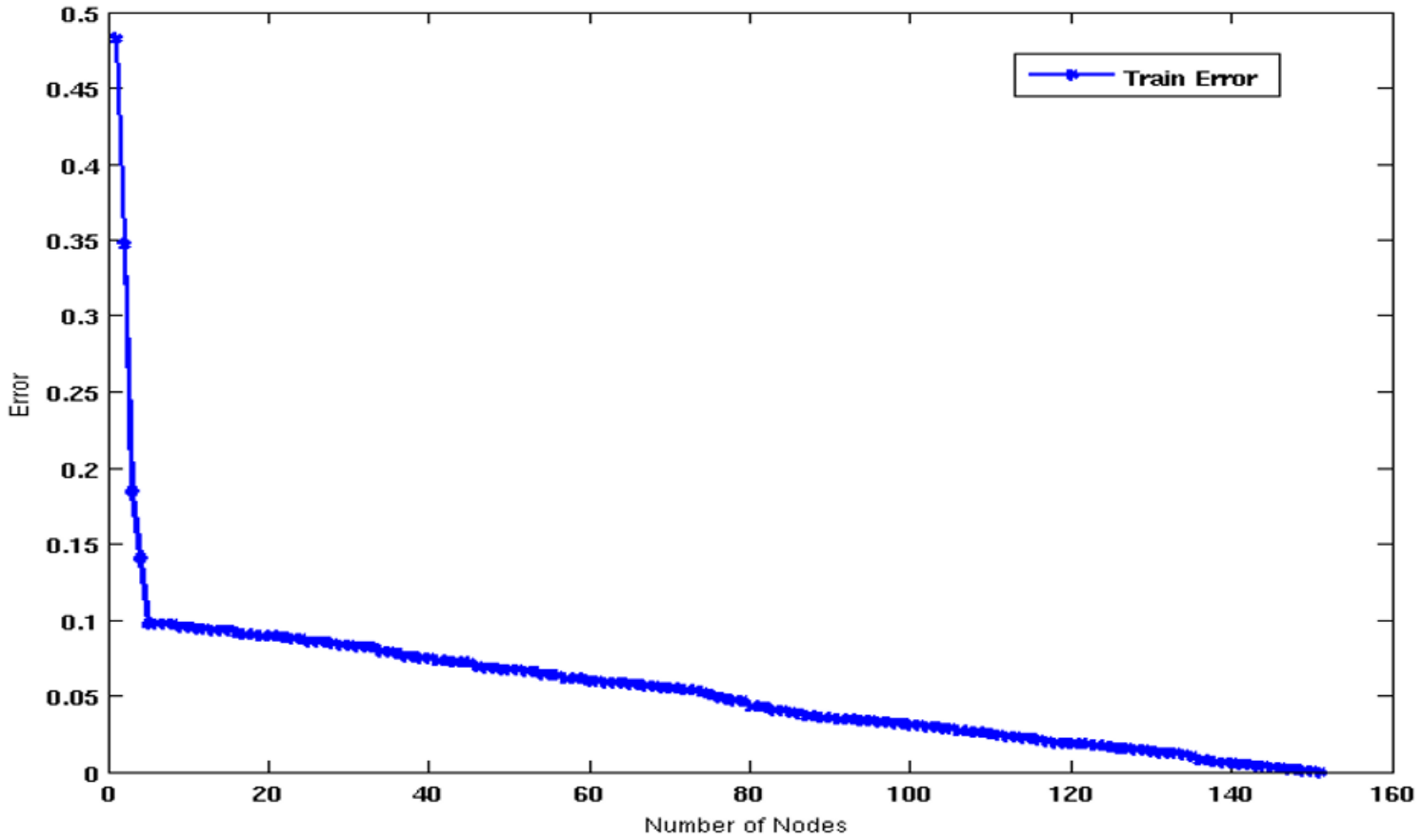
+ : 5400 instances

- 5000 instances generated from a Gaussian centered at (10,10)
- 400 noisy instances added

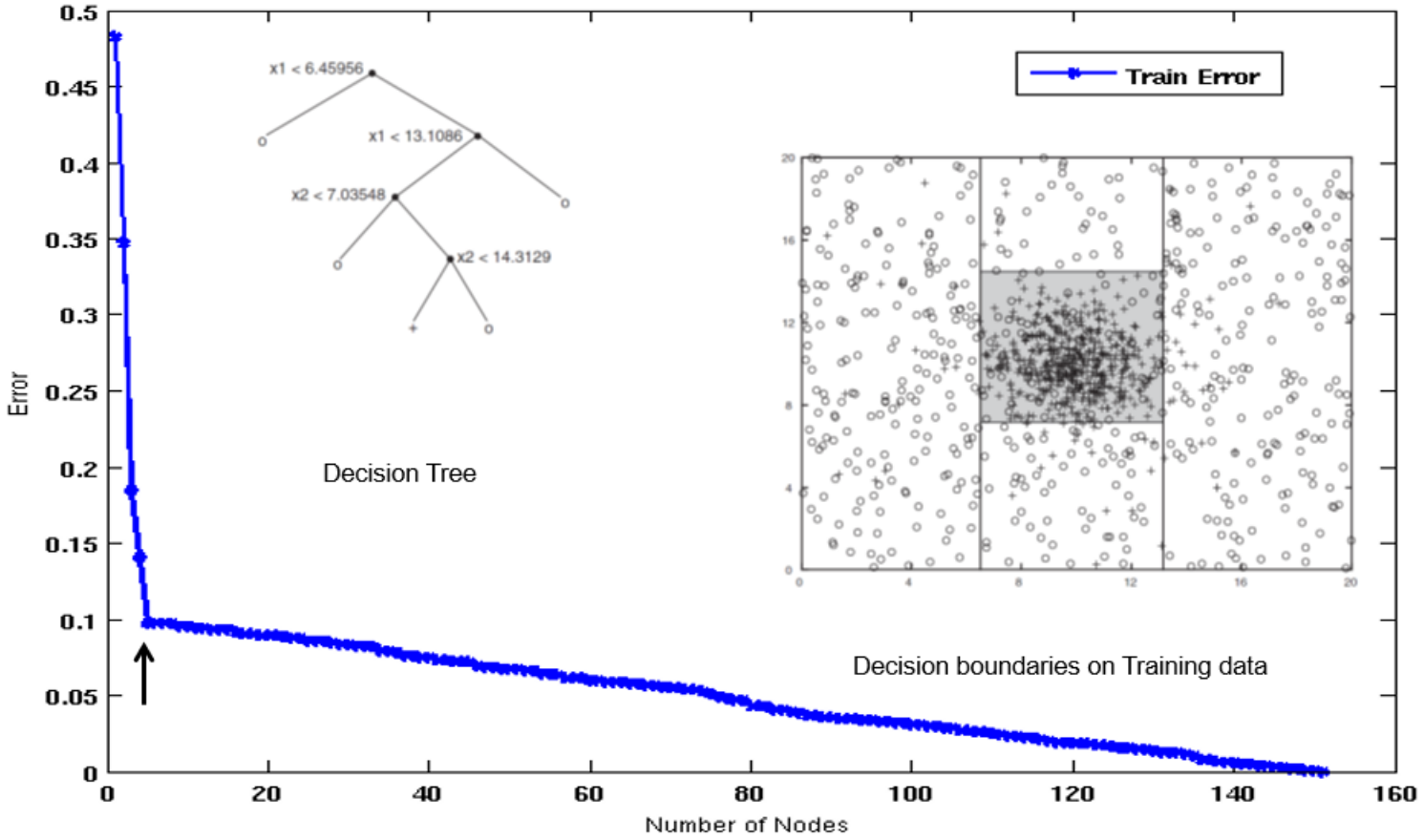
o : 5400 instances

- Generated from a uniform distribution

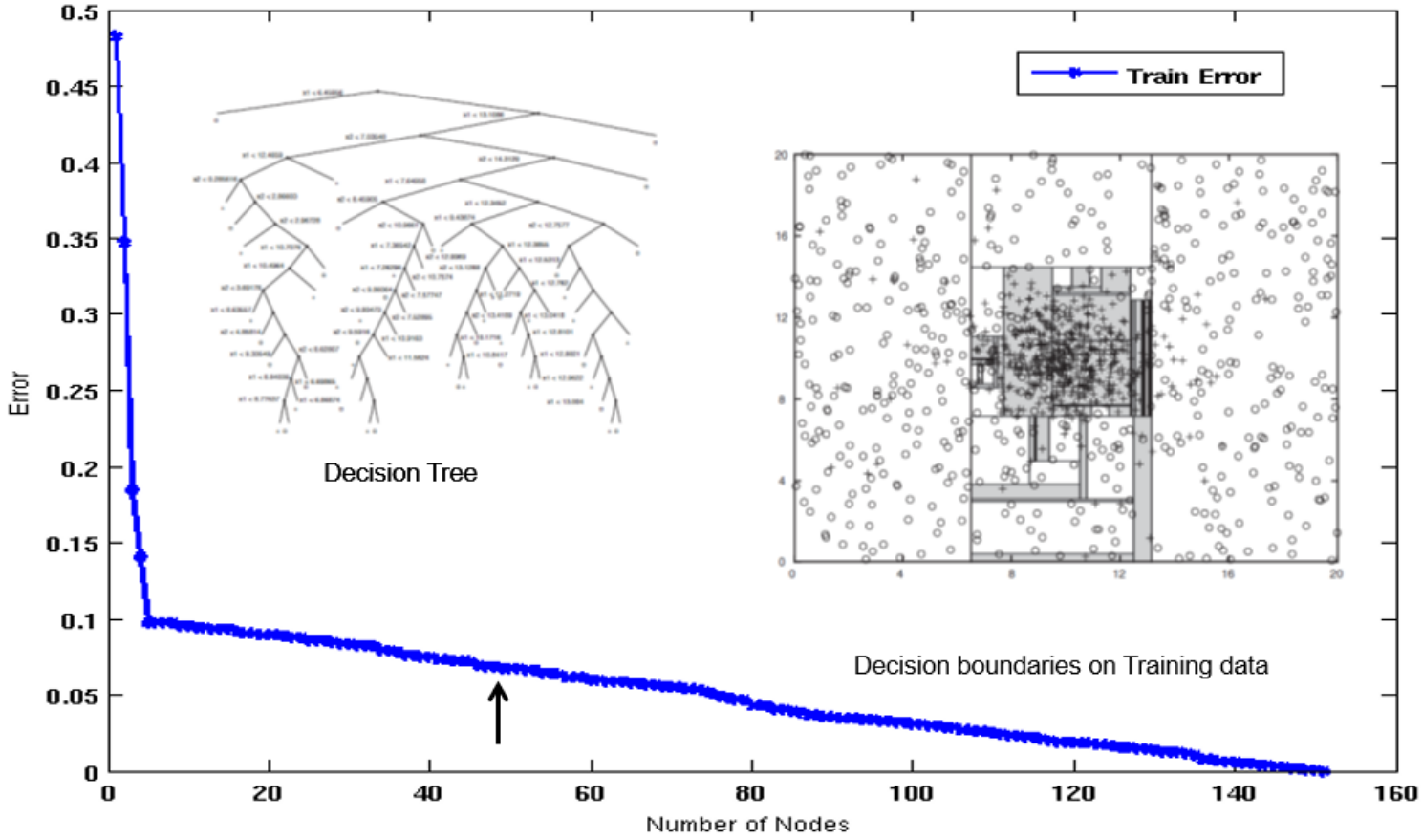
10 % of the data used for training and 90% of the data used for testing



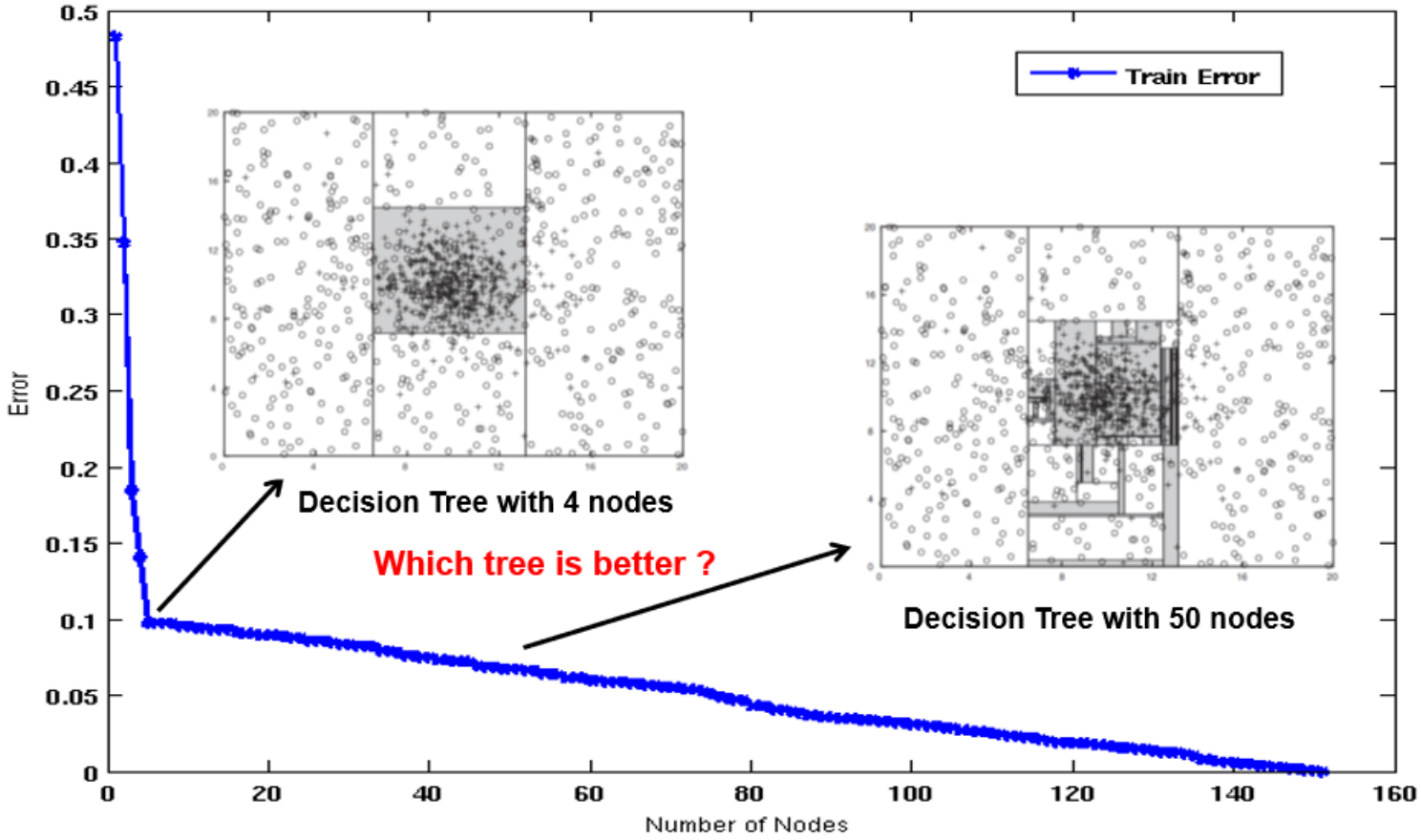
Increasing Number of Nodes in Decision Tree



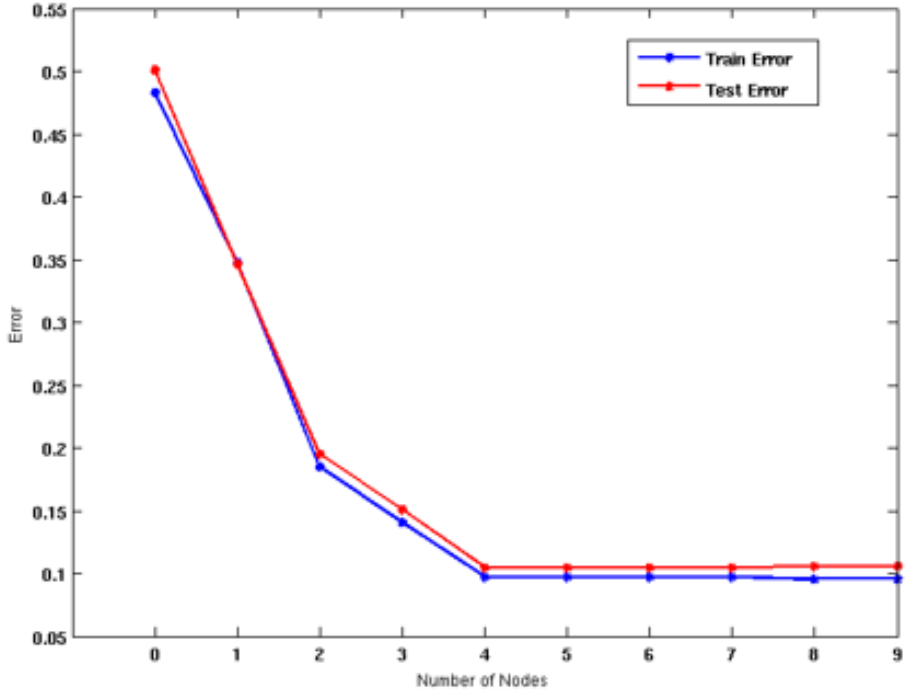
Decision Tree with 4 nodes

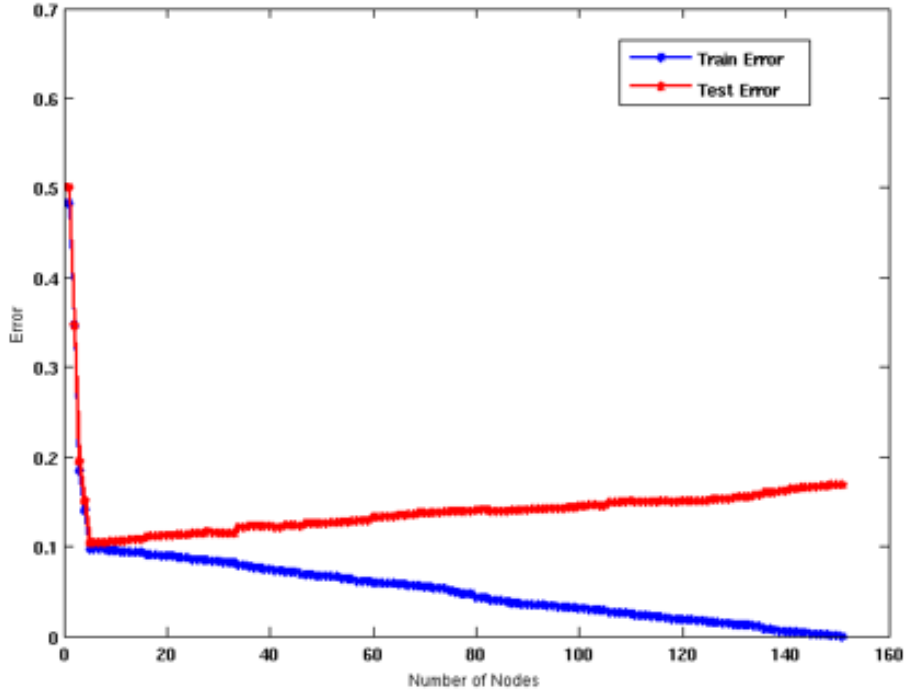


Decision Tree with 50 nodes



Decision Tree with 50 nodes



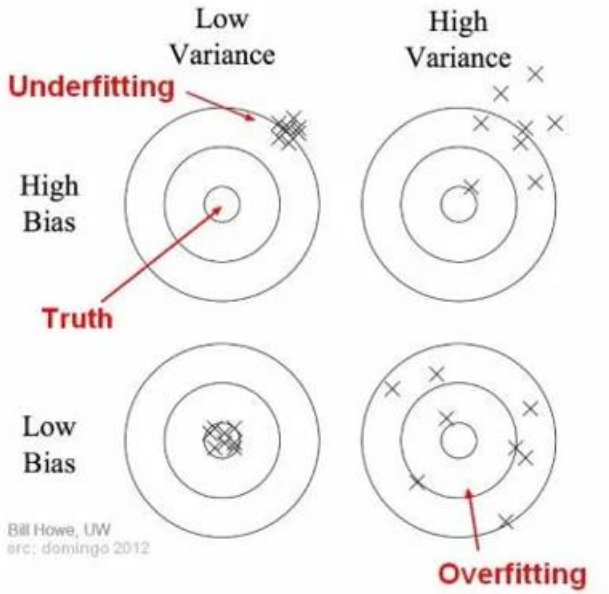


•As the model becomes more and more complex, test errors can start increasing even though training error may be decreasing

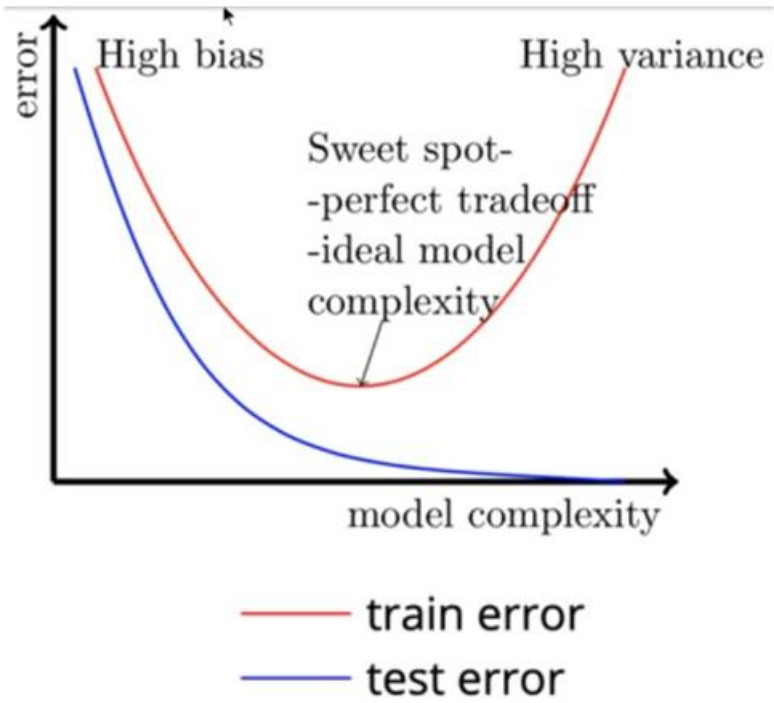
Model Underfitting & Model Overfitting

Underfitting: when model is too simple, both training and test errors are large

Overfitting: when model is too complex, training error is small but test error is large

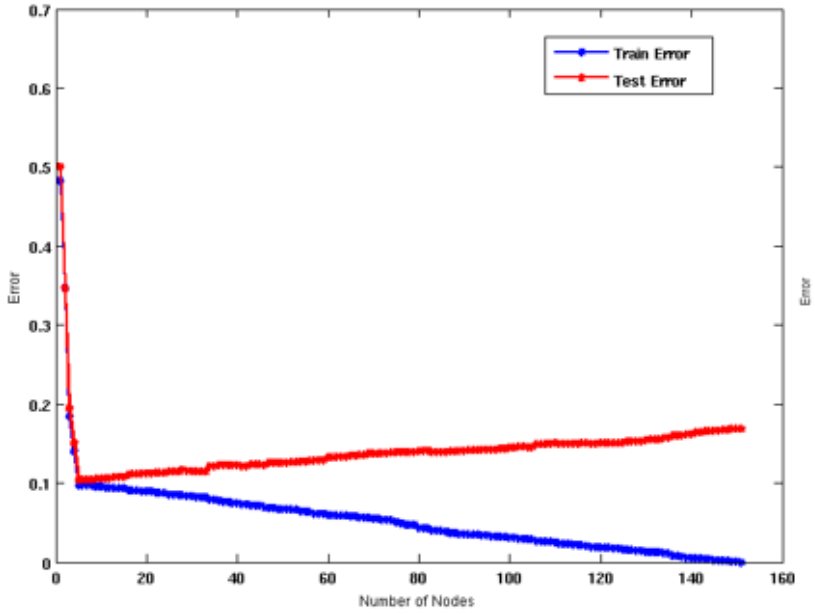


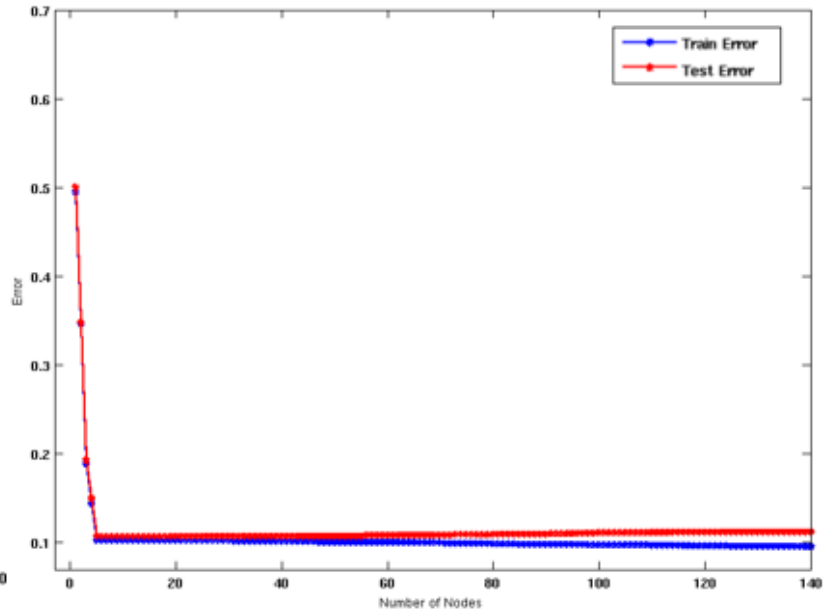
Simple Model: high bias, low variance
Complex Model: low bias, high variance
Ideal Model: low bias, low variance



Model Underfitting & Model Overfitting

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Overfitting: when model is too complex, training error is small but test error is large

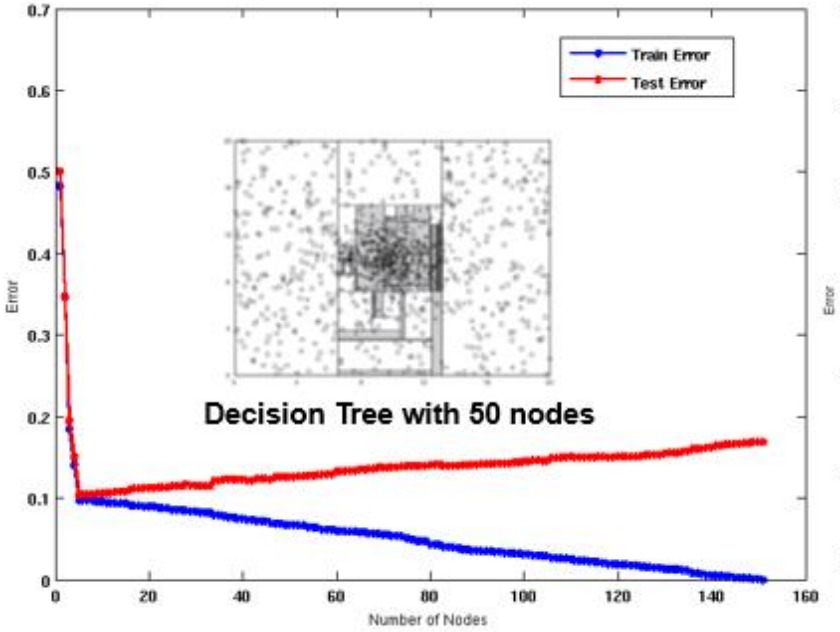


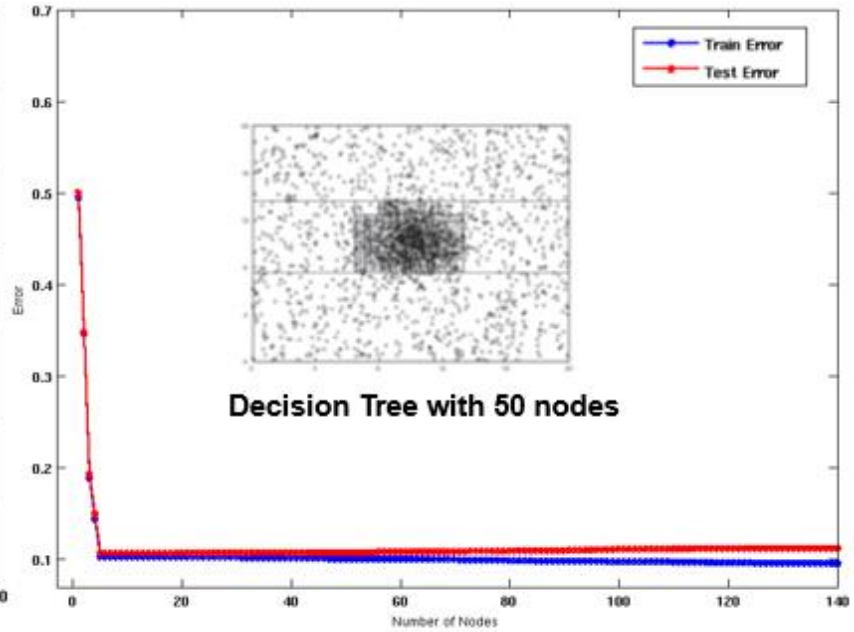


Using twice the number of data instances

- Increasing the size of training data reduces the difference between training and testing errors at a given size of model

Model Overfitting – Impact of Training Data Size





Using twice the number of data instances

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Model Overfitting – Impact of Training Data Size

- Not enough training data
- High model complexity
  - Multiple Comparison Procedure

Reasons for Model Overfitting

- Overfitting results in decision trees that are more complex than necessary
- Training error does not provide a good estimate of how well the tree will perform on previously unseen records
- Need ways for estimating generalization errors

Notes on Overfitting

- Divide training data into two parts:
  - Training set:
    - use for model building
  - Validation set:
    - use for estimating generalization error
    - Note: validation set is not the same as test set

- Drawback:
  - Less data available for training

Model Selection: Using Validation Set


thank
you